

EFFECTS OF DEVIATIONS FROM THE OPTIMUM ALLOCATION:-

Suppose that it is intended to use optimum allocation for given n . The sample size n_h in stratum h should be

$$n_h = \frac{n(W_h S_h)}{\sum W_h S_h} \quad \text{--- (1)}$$

From equation, the resulting minimum variance is

$$V_{\min}(\bar{y}_{st}) = \frac{1}{n} (\sum W_h S_h)^2 - \frac{1}{N} \sum W_h S_h^2 \quad \text{--- (2)}$$

In practice, since the S_h are not known, we can only approximate this allocation.

If $\hat{n}_h$ is the sample size used in stratum h , the variance actually attained,

$$V(\bar{y}_{st}) = \sum \frac{W_h^2 S_h^2}{\hat{n}_h} - \frac{1}{N} \sum W_h S_h^2$$

by replacing n_h by $\hat{n}_h$ in formula

$$V(\bar{y}_{st}) = \sum \frac{W_h^2 S_h^2}{\hat{n}_h} - \frac{1}{N} \sum W_h S_h^2$$

The increase in variance caused by the imperfect allocation is

$$V(\bar{y}_{st}) - V_{\min}(\bar{y}_{st}) = \sum \frac{W_h^2 S_h^2}{\hat{n}_h} - \frac{1}{N} \sum W_h S_h^2 - \frac{1}{n} (\sum W_h S_h)^2 + \frac{1}{N} \sum W_h S_h^2$$

$$= \sum \frac{W_h S_h^2}{\hat{n}_h} - \frac{1}{n} (\sum W_h S_h)^2 \quad \text{--- (3)}$$

On the first term on the right substitute for $W_h S_h$ in terms of n'_h from eq (1).
from equation: (1).

$$n'_h = \frac{n(W_h S_h)}{\sum W_h S_h}$$

$$W_h S_h = \frac{n'_h \sum W_h S_h}{n}$$

Put in eq (3), we have

$$\begin{aligned} V(\bar{y}_{st}) - V(\bar{y}_{st})_{\min} &= \sum \frac{(n'_h \frac{\sum W_h S_h}{n})^2}{\hat{n}_h} - \frac{1}{n} (\sum W_h S_h)^2 \\ &= \left(\frac{\sum W_h S_h}{n} \right)^2 \left[\sum \frac{n_h'^2}{\hat{n}_h} - n \right] \\ &= \left(\frac{\sum W_h S_h}{n} \right)^2 \left[\sum \frac{n_h'^2}{\hat{n}_h} - 2n + n \right] \\ &= \left(\frac{\sum W_h S_h}{n} \right)^2 \left[\sum \frac{n_h'^2}{\hat{n}_h} - 2 \sum n'_h + \sum \hat{n}_h \right] \\ &= \left(\frac{\sum W_h S_h}{n} \right)^2 \sum \frac{1}{\hat{n}_h} \left[n_h'^2 - 2 \hat{n}_h n'_h + \hat{n}_h^2 \right] \end{aligned}$$

$$V - V_{\min} = \frac{(\sum W_h S_h)^2}{n^2} \left[\sum \frac{(n'_h - \hat{n}_h)^2}{\hat{n}_h} \right] \quad \text{--- (4)}$$

Reverting to equation (2). If the fpc (last term on the right) is negligible, we see that

$$V_{\min} = \frac{(\sum W_h S_h)^2}{n}$$

Hence the proportional increase in variance resulting from deviations from the optimum allocation is

$$\begin{aligned} \frac{V - V_{\min}}{V_{\min}} &= \frac{(\sum W_h S_h)^2 \sum \frac{(\hat{n}_h - n'_h)^2}{\hat{n}_h}}{(\sum W_h S_h)^2} \\ &= \frac{1}{n} \sum_{h=1}^L \frac{(\hat{n}_h - n'_h)^2}{\hat{n}_h} \quad \text{--- (5)} \end{aligned}$$

where $\hat{n}_h$ = the actual sample size in stratum h .

And n'_h = the optimum sample size in stratum h .

If the fpc is not negligible, the equality sign in eq. (5) becomes ' $\geq$ '.

Let $g_h = \frac{|\hat{n}_h - n'_h|}{\hat{n}_h}$ be the absolute

difference in the sample size in stratum h , expressed as a fraction of the actual sample size $\hat{n}_h$. Then eq. (5) becomes

$$\frac{V - V_{\min}}{V_{\min}} = \sum_{h=1}^L \frac{\hat{n}_h}{n} g_h^2$$

a weighted mean of the g_h^2 .

A conservative upper limit to $(V - V_{\min}) / V_{\min}$

is therefore g^2 , where g is the largest proportional difference in any stratum.

Thus, if $g=0.2$ or 20%, the proportional increase cannot exceed $(0.2)^2$ or 40%.

If $g=30\%$ the proportional increase in variance is at most 9%.

THEOREM:- *

Given the results of a stratified random sample and unbiased estimates of $V(\bar{y}_{rand})$ from the same population is

$$V_{rand} = \hat{V}_{rand} = \frac{(N-n)}{n(N-1)} \left[\frac{1}{N} \sum \frac{N_h}{n_h} \sum_{i=1}^{n_h} y_{hi}^2 - \bar{y}_{st}^2 + v(\bar{y}_{st}) \right]$$

where $v(\bar{y}_{st})$ is the usual unbiased estimator of $V(\bar{y}_{st})$.

Proof:-

We know that

$$V_{rand} = \left(\frac{N-n}{N \cdot n} \right) S^2$$

$$\begin{aligned} \text{where } S^2 &= \frac{1}{N-1} \sum (y_i - \bar{y})^2 \\ &= \frac{1}{N-1} \left[\sum y_i^2 - \frac{(\sum y_i)^2}{N} \right] \end{aligned}$$

For stratified population

$$S^2 = \frac{1}{N-1} \left[\sum \sum y_{hi}^2 - \frac{(\sum \sum y_{hi})^2}{N} \right]$$

$$\text{So } V_{\text{rand}} = \frac{N-n}{N \cdot n} \left[\frac{1}{N-1} \left\{ \sum \sum y_{hi}^2 - \frac{(\sum \sum y_{hi})^2}{N} \right\} \right]$$

$$= \frac{N-n}{n(N-1)} \left[\frac{1}{N} \sum \sum y_{hi}^2 - \frac{(\sum \sum y_{hi})^2}{N^2} \right]$$

$$\text{and } V_{\text{rand}} = \frac{N-n}{n(N-1)} \left[\frac{1}{N} \sum \sum y_{hi}^2 - \bar{Y}^2 \right] \quad (\because \bar{Y} = \frac{\sum \sum y_{hi}}{N}) \quad \text{--- (I)}$$

$$\text{Now } \text{Est}(V_{\text{rand}}) = \frac{N-n}{n(N-1)} \left[\text{Est} \left(\frac{1}{N} \sum \sum y_{hi}^2 \right) - \text{Est} \bar{Y}^2 \right] \quad \text{--- (I)}$$

$$\frac{1}{N} E \left[\sum_h \frac{N_h}{n_h} \sum_i^{n_h} y_{hi}^2 \right] = \frac{1}{N} \sum_{h=1}^L \frac{N_h}{n_h} \sum_i^{n_h} y_{hi}^2 = \frac{1}{N} \sum_{h=1}^L \sum_i^{n_h} y_{hi}^2$$

$$\text{Est} \left[\frac{1}{N} \sum_h \frac{N_h}{n_h} \sum_i^{n_h} y_{hi}^2 \right] = \frac{1}{N} \sum_h \frac{N_h}{n_h} \sum_i^{n_h} y_{hi}^2 \quad \text{--- (II)}$$

Since $v(\bar{y}_{st})$ and $\bar{y}_{st}$ are unbiased estimates of $V(\bar{y}_{st})$ and $\bar{Y}$ respectively.

$$\begin{aligned} E[v(\bar{y}_{st})] &= V(\bar{y}_{st}) \\ &= E[\bar{y}_{st} - \bar{Y}]^2 \\ &= E(\bar{y}_{st}^2) - [E(\bar{y}_{st})]^2 \quad \because \bar{Y} = E(\bar{y}_{st}) \\ &= E(\bar{y}_{st}^2) - \bar{Y}^2 \end{aligned}$$

$$\bar{Y}^2 = E(\bar{y}_{st}^2) - E[v(\bar{y}_{st})]$$

$$\text{Est}(\bar{Y}^2) = \bar{y}_{st}^2 - v(\bar{y}_{st}) \quad \text{--- (III)}$$

And since $E(V_{\text{rand}}) = V_{\text{rand}}$ so $\text{Est}(V_{\text{rand}}) = V_{\text{rand}}$

Putting the results from eq (II) & (III) in (I).

$$\text{Est } V_{\text{rand}} = \frac{N-n}{(N-1)n} \left[\frac{1}{N} \sum_h \frac{N_h}{n_h} \sum_i^{n_h} y_{hi}^2 - \bar{y}_{st}^2 + v(\bar{y}_{st}) \right]$$

$$V_{\text{rand}} = \frac{N-n}{(N-1)n} \left[\frac{1}{N} \sum_h \frac{N_h}{n_h} \sum_i^{n_h} y_{hi}^2 - \bar{y}_{st}^2 + v(\bar{y}_{st}) \right] \quad \text{--- (IV)}$$

Proved

With proportional allocation: $(N_h/n_h = N/n)$;

The first two terms inside the square brackets becomes $(\frac{1}{n})$ times the within

sample sum of squares = $(n-1) S^2$

$$[\because S^2 = \frac{1}{n-1} \sum y_i^2 - \frac{(\sum y_i)^2}{n}]$$

Then the expression (4) reduces to

$$V_{\text{rand}} = \frac{N-n}{(N-1)n} \left[\frac{(n-1) S^2}{n} + V(\bar{y}_{st}) \right] \quad (5)$$

If n is large $(n-1) = n$ and $N-1 = N$ and the term in $V(\bar{y}_{st})$ is of order $\frac{1}{n}$ relative to the term in S^2 .

So, Eq (5) reduces to

$$V_{\text{rand}} = \frac{N-n}{n \cdot N} S^2$$

It is just for proportional allocation. In the general case the corresponding simplification (n large) is

$$V_{\text{rand}} = \frac{N-n}{n \cdot N} \left[\frac{1}{N} \sum \frac{N_h}{n_h} \sum y_{hi}^2 - \bar{y}_{st}^2 \right]$$

TWO WAY STRATIFICATION:

(With Small sample size)

Suppose that there are two criteria of stratification say 'R' rows and 'C' columns making RC cells. If $n \geq RC$ every cell can be represented in the sample. A problem arises when $n < RC$ and we would like the sample to give proportional representation to each criterion of stratification.

In a simple method developed by Bryant, Hartely & Jessen. The techniques required only that n exceeds the greater of R or C. To illustrate this method:

Suppose we have small population of 165 schools has been stratified by size of city into 3 classes and by average expenditures per student into 4 classes. The no of schools in each cell is denoted by m_{ij} and the proportion of school is P_{ij} (proportion of the i th and j th cell).

$$\text{where } P_{ij} = \frac{m_{ij}}{N}$$

Criteria I

	A	B	C	D	$m_{i.}$ $= P_{i.}$	$n_{i.}$
I					$m_{1.}$ $= P_{1.}$	$n_{1.} = 4$
II					$m_{2.}$ $= P_{2.}$	$n_{2.} = 2$
III					$m_{3.}$ $= P_{3.}$	$n_{3.} = 2$
$m_{.j}$	$m_{.1}$	$m_{.2}$	$m_{.3}$	$m_{.4}$	N	$\downarrow$
$n_{.j}$	$n_{.1} = 2$	$n_{.2} = 2$	$n_{.3} = 2$	$n_{.4} = 2$	$\longrightarrow n = 8$	

$$P_{i.} = \frac{m_{i.}}{N}, \quad P_{.j} = \frac{m_{.j}}{N}$$

$$n_{i.} = n P_{i.}, \quad n_{.j} = n P_{.j}$$

$$\sum n_{i.} = n, \quad \sum n_{.j} = n.$$

And $m_{.j} = m_{i.} =$ margin stratum size.

The objective is to give each school an approximately equal chance of selection while giving each marginal class its proportional representation. In this illustration, let say $n=8$ compute the $n_{i.} = n P_{i.}$ and $n_{.j} = n P_{.j}$ where these values are rounded to the nearest integers.

It should be further noted while making the adjustment that $n_{i.}$ and $n_{.j}$ both add to n .

The next step is to draw $n=8$ cells with Probabilities

$$P_i \times P_j$$

$$\frac{n_{i.}}{n} \times \frac{n_{.j}}{n} = \frac{(n_{i.})(n_{.j})}{n^2}$$

for the i th and j th cell.
This is done by constructing an $n \times n$ i.e. (8×8) .

	A	B	C	D
1	x			
2				
3		x		
4			x	
1			x	
2				x
1		x		
2				x

In row I one column is drawn at random. In row II one of the remaining column is drawn at random and so at the end each row and column contains 1 unit. The result of one draw are indicated by (x) cross.

Similarly rows 1 through 4 are assigned to margin stratum I and so on. This complete the allocation

of sample 2 the 12 cells are

	A	B	C	D	Total
I	2	1	1	0	4
II	0	0	1	1	2
III	0	1	0	1	2
Total	2	2	2	2	→ 8

An unbiased estimator of mean is

$$\bar{y}_u = \frac{1}{n} \sum_{ij} \frac{n_i^2}{n_i \cdot n_j} p_{ij} y_{ij}$$

where y_{ij} is the sample total in the i th and j th cell. A sample estimate of variance is available for this $(\bar{y}_u)$ estimate. We can have the variance and the estimator variance of $\bar{y}_u$ also.

CONTROLLED SELECTION:-

Goodman and Kish (1950) suggested a procedure of selection called controlled selection in these situations in which it is considered desirable that certain combinations of units are preferred combinations be given greater probability of selection and less probability of selection for some or all non-preferred combinations.

Example: Suppose in a block growing field. It is proposed to select a simple random sample of size 2 fields to estimate the average yield per hectare. In the sample there are 3 probs.

1. Both fields are irrigated.
2. Both fields are un-irrigated.
3. One is irrigated and other is un-irrigated.

Undoubtedly all samples will give an unbiased estimate of the average yield but it is likely that the first two samples will be overestimating or underestimating, while in the last case it is likely to be much

class to the parameter. Hence, it is desirable to group the fields into two strata, one which irrigated fields and the other with un-irrigated and then to select a simple random sample of one field from each stratum. Such samples are called preferred sample and the others may be called non-preferred sample. At this stage it may be mentioned that the probability of some or all preferred combinations of units are increased and those of some or all non-preferred combinations are reduced.

Thus Stratified random sampling can be considered a method of controlled selection. However in this section the term controlled selection is used in the sense as having additional controls in stratification for reducing variance for the estimate of the parameter under study.

METHOD OF COLLAPSED STRATA:-

If the population is highly heterogeneous and several criteria are available for stratification, it may be desirable to stratify population to the extent that only one sample can be drawn from each stratum. When this is the case, it is not possible to estimate the variability within each stratum and consequently the variances of the estimated field. In such cases an appropriate estimate of the variance of estimated mean is usually obtained by using the method of collapsed stratum. The method consists in grouping together pairs of strata whose population mean $\bar{Y}_h$ do not differ very much from each other.

POST STRATIFICATION:-

THE CONSTRUCTION OF THE STRATA:-

The remarks for the construction of a strata are discussed below.

D. Classifying Sampling units:-

Every sampling unit must be classified distinctly into one of the strata i.e. information must be available on all of the sampling units in the population. Information available for only a small part of the sampling units is not generally useful for stratification. This strict rule of universal information can be relaxed in several ways.

2). Stratification with Random "Quotas":-

If the stratifying variable and stratum weights W_h are also known, then the desired sample size n_h should be specified and filled by drawing selections at random from the entire population.

The ' n_h ' selection from the h th stratum until the 'quotas' of n_h are filled represent a random selection.

Errors in Sorting:-

If after selection a few units are discovered to have been sorted into the "wrong" strata is generally best to leave them in their sorted strata. This will merely decrease slightly the efficiency of the stratification. But it will not bias the selection, because on the average the sample will contain merely its portion of wrong sorting.

Overlapping Lists:-

If the population frame is defined as the sum of several lists from each of which a sample is selected separately, then each list becomes a stratum. The appearance of some sampling units on two or more lists results in duplicate listings.

Objectivity and Regularity Unnecessary:-

Neither objectivity nor regularity are needed for sorting sampling units into strata.

- 1). On the contrary, subjective sorting may be superior to rigid procedures for creating homogeneous strata.
- 2). Regularity is not required, For one thing the limits of class boundaries may be changed and for another the classes need not be used symmetrically.

6). Homogeneity within Strata:

For smaller variance, we need stratifying variables closely related to the main survey objectives. The sampling units are relatively homogeneous in survey variables. Their variance are reduced to the extent that the variation among sampling units within the strata is less than their variation in the entire population. We try to increase and maximize the homogeneity of the sampling units within strata.

7). Utilizing available variables:-

It may not be clear

just which variables $\rightarrow$ yield more gain. Several practical hints are in order.

Generally, more gain accrues from the use of coarser divisions of several variables than from the finer division of one.

When using several variables, there is no need for completeness and symmetry in forming the cells; smaller and less important cells may be combined.

Stratifying variables unrelated to each other should be preferred. Contrariwise, if two stratifying variables are highly correlated, using either one will give as much gain as using both.

Elaborate Controls Unnecessary:-

Elaborate controls for several stratifying variables would result in too many strata. The number of strata must be limited by the effort that can be profitably spent on stratification. At least two sampling units must be selected from each

stratum to compute an unbiased estimate of the variance. Hence the number of strata should not be greater than half the number of sampling units the study can afford.

9). Number of Strata:-

Another problem is to find that how many strata should be formed from any single stratifying variable. Very small strata contribute little to the gains from stratification. Perhaps between three and ten strata suffice for any single variable.

SYSTEMATIC SAMPLING

In Systematic Sampling, the population of 'N' units is numbered from 1 to N. Then Population is divided into k units. To select a sample of size 'n', we take a unit at random between 1 and 'k' units and every kth unit is selected thereafter.

For example:- of population of size $N=16$, $K=4$,

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16.

$$n = \frac{N}{K} = \frac{16}{4} = 4.$$

Suppose from the first 'K' units '3' is selected, then from the next 'K' units selected unit would be 7 and 11, 15 from remaining.

i.e. $3+k$ th, $3+2k$ th, $3+3k$ th.

The selection of first unit determines the whole sample.

Here 'N' is an integral multiple of n.

And 'K' is the no of all possible samples when we are doing the Systematic Sampling.

Another method of choosing each unit is that, instead of starting from random numbers chosen between 1 and K , we take starting numbers as $\frac{K+1}{2}$ if K is odd and $\frac{K+2}{2}$ or $\frac{K}{2}$ th value for K is even.

If $\frac{K}{2}$ th unit is the first unit selected in the sample then ' K ' is even.

If $\frac{K+1}{2}$ th unit is first unit selected in the sample then K is odd and these are Centrally located sample.

Systematic Sample numbers

I	II	III	IV	V
1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	

$$N=23, K=5$$

The first three samples have $n=5$ and last two have $n=4$. This fact introduces a disturbance into the theory of systematic sampling. The disturbance is negligible if $n > 50$.

It is unlikely to be large even when n is small.

Circular Systematic Sampling:-

A method suggested by Lahiri is that select a random number between 1 and N and take every k th unit going round the circle until desired sample n is obtained.

In this sample, we have

$$N=23, k=5$$

We want $n=5$

Suppose we randomly select number 16, then we take every unit with the difference of 5 and going on circular as 16, 21, 3, 8, 13.

If $n=5$, units are wanted with $N=23$, we take $k=6$.

Notation:-

$\bar{y}_i$ = The Sample mean of i th systematic

$$\text{Sample} = \frac{\sum_{j=1}^n y_{ij}}{n}$$

$$\bar{y} = \text{overall mean} = \frac{\sum_{i=1}^N \bar{y}_i}{N} = \frac{\sum_{i=1}^N \sum_{j=1}^n y_{ij}}{N}$$

Let $y_{i.} = \sum_j y_{ij}$ = the total for i th Sys-Samp.

$$y_{i.} = n \bar{y}_{i.}$$

$$\bar{y}_{i.} = \frac{\sum_j y_{ij}}{n} = \frac{y_{i.}}{n}$$

$$\bar{Y} = \frac{\sum_i y_{i.}}{N} = n \frac{\sum_i \bar{y}_{i.}}{nk} = \frac{1}{k} \sum_i \bar{y}_{i.}$$

Variance:-

$\bar{y}_{sys}$ = overall mean of Systematic Sample.
= overall estimate of the population mean from Sys Sampling.

$V(\bar{y}_{sys})$ = The variance of the mean of a Systematic Sample.

$$\begin{aligned} S_{wsy}^2 &= \text{variance within Systematic Sample.} \\ &= \frac{\sum_i S_{wi}^2}{k} = \frac{\sum_i \sum_j (y_{ij} - \bar{y}_{i.})^2}{k(n-1)} \end{aligned}$$

S_{wsy}^2 = Average of all variance within the i th Systematic Sample.

$$\begin{aligned} S_{wi}^2 &= \text{variance within } i\text{th Systematic Sample.} \\ &= \frac{\sum_j (y_{ij} - \bar{y}_{i.})^2}{n-1} \end{aligned}$$

$$S^2 = \frac{\sum_i \sum_j (y_{ij} - \bar{Y})^2}{N-1}$$

But

$$S_{wsy}^2 = \frac{1}{k(n-1)} \sum_i^k \sum_j^n (y_{ij} - \bar{y}_{i.})^2$$

$$\sum_i \sum_j (y_{ij} - \bar{y}_{i.})^2 = k(n-1) S_{wsy}^2 \quad \text{--- (2)}$$

Now by definition,

$$V(\bar{y}_{sys}) = \frac{1}{k} \sum_i^k (\bar{y}_{i.} - \bar{Y})^2$$

$$\sum_i^k (\bar{y}_{i.} - \bar{Y})^2 = k V(\bar{y}_{sys}) \quad \text{--- (3)}$$

put the values of eq (2) & (3) in (1).

$$\begin{aligned} (N-1)S^2 &= \sum_i \sum_j (y_{ij} - \bar{y}_{i.})^2 + n \sum_i (\bar{y}_{i.} - \bar{Y})^2 \\ &= k(n-1) S_{wsy}^2 + nk V(\bar{y}_{sys}). \end{aligned}$$

$$nk V(\bar{y}_{sys}) = (N-1)S^2 - k(n-1) S_{wsy}^2$$

$$N V(\bar{y}_{sys}) = (N-1)S^2 - k(n-1) S_{wsy}^2$$

$$V(\bar{y}_{sys}) = \left(\frac{N-1}{N} \right) S^2 - \frac{k(n-1)}{N} S_{wsy}^2$$

Proved.

THEOREM:- ✓

Show that the mean of Systematic Sample is more precise than the mean of a S.R.S if and only if

$$S_{wsy}^2 > S^2.$$

Proof:-

If mean of a Systematic Sampling is to be more precise than the Simple Random Sampling, then

THEOREM:-

The variance of the mean of a systematic sample is given by

$$V(\bar{y}_{sys}) = \left(\frac{N-1}{N}\right) S^2 - \frac{k(n-1)}{N} S_{wsys}^2$$

where S_{wsys}^2 = Variance within systematic sample.

Proof:-

By using the usual identity of Analysis of Variance

$$(N-1)S^2 = \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y})^2$$

Adding and Subtracting $\bar{y}_{i.}$ within brackets.

$$\begin{aligned}(N-1)S^2 &= \sum_{i=1}^k \sum_{j=1}^n [y_{ij} - \bar{y}_{i.} + \bar{y}_{i.} - \bar{y}]^2 \\ &= \sum_{i=1}^k \sum_{j=1}^n [(y_{ij} - \bar{y}_{i.}) + (\bar{y}_{i.} - \bar{y})]^2\end{aligned}$$

$$= \sum_{i=1}^k \sum_{j=1}^n \left[(y_{ij} - \bar{y}_{i.})^2 + (\bar{y}_{i.} - \bar{y})^2 + 2(y_{ij} - \bar{y}_{i.})(\bar{y}_{i.} - \bar{y}) \right]$$

Applying $\sum_{j=1}^n$ within the brackets.

$$(N-1)S^2 = \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2 + n \sum_{i=1}^k (\bar{y}_{i.} - \bar{y})^2 + 0.$$

And cross product terms become zero

$$\text{i.e. } \sum_{j=1}^n (y_{ij} - \bar{y}_{i.}) = 0 \quad \text{or} \quad \sum_{i=1}^k (\bar{y}_{i.} - \bar{y}) = 0.$$

So

$$(N-1)S^2 = \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2 + n \sum_{i=1}^k (\bar{y}_{i.} - \bar{y})^2 \quad \text{--- (1)}$$

$$V(\bar{y}_{sys}) < V(\bar{y}_{srs})$$

$$\left(\frac{N-1}{N}\right) S^2 - \frac{K(n-1)}{N} S_{wsy}^2 < \left(\frac{N-n}{N \cdot n}\right) S^2$$

$$\frac{1}{N} [(N-1)S^2 - K(n-1)S_{wsy}^2] < \frac{N-n}{N \cdot n} S^2$$

$$-K(n-1)S_{wsy}^2 < \left(\frac{N-n}{n} - N+1\right) S^2$$

$$-K(n-1)S_{wsy}^2 < \left(\frac{N-n - nN + n}{n}\right) S^2$$

$$-nK(n-1)S_{wsy}^2 < -N(n-1)S^2$$

$$-N(n-1)S_{wsy}^2 < -N(n-1)S^2$$

$$S_{wsy}^2 > S^2$$

Systematic Sample is more precise when the units within Systematic Sample are heterogeneous, and is imprecise when the units within Systematic Samples are homogeneous.

THEOREM:-

Show that

$$V(\bar{y}_{sys}) = \frac{S^2}{n} \left(\frac{N-1}{N}\right) [1 + (n-1)\rho_w]$$

where ρ_w is the correlation coefficient between pairs of units that lie in the same systematic sample

$$\rho_w = \frac{E(y_{ij} - \bar{y})(y_{ik} - \bar{y})}{E(y_{ij} - \bar{y})^2}$$

$$S^2 = \frac{\sum (y_i - \bar{Y})^2}{N-1}$$

Proof:-

By definition

$$V(\bar{y}_{sys}) = \frac{1}{K} \sum_i^k (\bar{y}_i - \bar{Y})^2$$

$$\begin{aligned} K V(\bar{y}_{sys}) &= \sum_i^k (\bar{y}_i - \bar{Y})^2 \\ &= \sum_i^k \left[\sum_j^k \frac{y_{ij}}{n} - \bar{Y} \right]^2 = \sum_i^k \left[\frac{1}{n} \left(\sum_j^k y_{ij} - n\bar{Y} \right) \right]^2 \\ &= \sum_i^k \left[\frac{1}{n} \sum_j^k (y_{ij} - \bar{Y}) \right]^2 \end{aligned}$$

$$= \frac{1}{n^2} \sum_i^k \left[\sum_j^k (y_{ij} - \bar{Y})^2 + \sum_{j \neq u} (y_{ij} - \bar{Y})(y_{iu} - \bar{Y}) \right]$$

$$= \frac{1}{n^2} \left[\sum_i^k \sum_j^k (y_{ij} - \bar{Y})^2 + \sum_i^k \sum_{j \neq u} (y_{ij} - \bar{Y})(y_{iu} - \bar{Y}) \right]$$

As we have

$$S^2 = \frac{\sum_i^k \sum_j^n (y_{ij} - \bar{Y})^2}{N-1}$$

$$\sum_i^k \sum_j^n (y_{ij} - \bar{Y})^2 = (N-1) S^2 \quad \text{--- (2)}$$

We define

$$\rho_w = \frac{E(y_{ij} - \bar{Y})(y_{iu} - \bar{Y}) / nK(n-1)}{E(y_{ij} - \bar{Y})^2 / nK}$$

$$E(y_{ij} - \bar{Y})(y_{iu} - \bar{Y}) = \frac{\sum_i^k \sum_{j \neq u} (y_{ij} - \bar{Y})(y_{iu} - \bar{Y})}{K n(n-1)}$$

$$\text{And } E(y_{ij} - \bar{Y})^2 = \frac{\sum_i^k \sum_j^n (y_{ij} - \bar{Y})^2}{nK} = \frac{\sum_i^k \sum_j^n (y_{ij} - \bar{Y})^2}{nK} = \frac{(N-1) S^2}{N}$$

Dr. Ummeera Anwar H. Bhutto Sir

$$e_w = \frac{\sum_j \sum_u (y_{ij} - \bar{y})(y_{iu} - \bar{y})}{nK(n-1)} \quad (\because nK = N)$$

$$= \frac{\left(\frac{N-1}{N}\right) S^2}{(n-1)(N-1) S^2} \sum_j \sum_u (y_{ij} - \bar{y})(y_{iu} - \bar{y})$$

$$\sum_j \sum_u (y_{ij} - \bar{y})(y_{iu} - \bar{y}) = (n-1)(N-1) S^2 e_w. \quad \text{--- (B)}$$

Now putting the results from eq (2) & (3) in eq (1), we get

$$KV(\bar{y}_{sys}) = \frac{1}{n^2} [N-1 S^2 + (n-1)(N-1) S^2 e_w]$$

$$= \left(\frac{N-1}{n^2}\right) S^2 [1 + (n-1) e_w]$$

$$V(\bar{y}_{sys}) = \frac{N-1}{n^2 K} S^2 [1 + (n-1) e_w]$$

$$= \frac{S^2}{n} \cdot \frac{N-1}{nK} [1 + (n-1) e_w]$$

$$V(\bar{y}_{sys}) = \frac{S^2}{n} \cdot \left(\frac{N-1}{N}\right) [1 + (n-1) e_w]$$

Ans req.

This result shows that +ve correlation between units in the same sample inflates the variance of the sample mean. Even a small positive correlation may have a large effect because of the multiplier $(n-1)$.

THEOREM:-

Show that

$$V(\bar{y}_{sys}) = \frac{(N-n)}{N} \frac{S_{wst}^2}{n} [1 + (n-1) \rho_{wst}]$$

where $S_{wst}^2 = \frac{1}{n(k-1)} \sum_j^n \sum_i^k (y_{ij} - \bar{y}_{.j})^2$ is the

variance among the units that lie in the same stratum, the divisor $n(k-1)$ is used because n strata contribute $(k-1)$ d.f.

$$\rho_{wst} = \frac{E(y_{ij} - \bar{y}_{.j})(y_{iu} - \bar{y}_{.u})}{E(y_{ij} - \bar{y}_{.j})^2}$$

which is the correlation coefficient between the pairs of deviations from the stratum means for the values that lie in the same systematic sample.

Proof:-

By definition

$$V(\bar{y}_{sy}) = \frac{1}{k} \sum_i^k (\bar{y}_{i.} - \bar{Y})^2$$

$$\bar{y}_{i.}$$

$$= \frac{1}{k} \sum_i^k \left[\sum_j^n \frac{y_{ij}}{n} - \sum_j^n \frac{\bar{y}_{.j}}{n} \right]^2$$

$$= \frac{1}{k} \sum_i^k \left[\frac{1}{n} \sum_j^n (y_{ij} - \bar{y}_{.j}) \right]^2$$

$$\begin{aligned}
 V(\bar{y}_{sys}) &= \frac{1}{n^2 k} \sum_{j=1}^k \left[\sum_{i=1}^n (y_{ij} - \bar{y}_{.j})^2 \right] \\
 &= \frac{1}{n^2 k} \sum_{j=1}^k \left[\sum_{i=1}^n (y_{ij} - \bar{y}_{.j})^2 + 2 \sum_{j < u} (\bar{y}_{.j} - \bar{y}_{.u}) (\bar{y}_{.j} - \bar{y}_{.u}) \right] \\
 &= \frac{1}{n^2 k} \left[\sum_{j=1}^k \sum_{i=1}^n (y_{ij} - \bar{y}_{.j})^2 + 2 \sum_{j < u} (\bar{y}_{.j} - \bar{y}_{.u}) (\bar{y}_{.j} - \bar{y}_{.u}) \right]
 \end{aligned}$$

As $S_{wst}^2 = \frac{1}{n(k-1)} \sum_{j=1}^k \sum_{i=1}^n (y_{ij} - \bar{y}_{.j})^2$

$$\sum_{j=1}^k \sum_{i=1}^n (y_{ij} - \bar{y}_{.j})^2 = n(k-1) S_{wst}^2$$

So

$$V(\bar{y}_{sys}) = \frac{1}{n^2 k} \left[n(k-1) S_{wst}^2 + 2 \sum_{j < u} (\bar{y}_{.j} - \bar{y}_{.u}) (\bar{y}_{.j} - \bar{y}_{.u}) \right] \quad \text{--- (1)}$$

Now $e_{wst} = \frac{E(\bar{y}_{.j} - \bar{y}_{.u}) (\bar{y}_{.j} - \bar{y}_{.u})}{E(\bar{y}_{.j} - \bar{y}_{.j})^2}$

$$E(\bar{y}_{.j} - \bar{y}_{.u}) (\bar{y}_{.j} - \bar{y}_{.u}) = \frac{\sum_{j < u} (\bar{y}_{.j} - \bar{y}_{.u}) (\bar{y}_{.j} - \bar{y}_{.u})}{n k (n-1)}$$

And $E(\bar{y}_{.j} - \bar{y}_{.j})^2 = \frac{\sum_{j=1}^k (\bar{y}_{.j} - \bar{y}_{.j})^2}{k \cdot n \cdot 1} = \frac{S_{wst}^2}{k \cdot n \cdot 1}$

$$e_{wst} = \frac{2 \sum_{j < u} (\bar{y}_{.j} - \bar{y}_{.u}) (\bar{y}_{.j} - \bar{y}_{.u})}{n k (n-1) \cdot \frac{\sum_{j=1}^k (\bar{y}_{.j} - \bar{y}_{.j})^2}{k \cdot n \cdot 1}}$$

$$e_{wst} = \frac{2 \sum_{j < u} (\bar{y}_{.j} - \bar{y}_{.u}) (\bar{y}_{.j} - \bar{y}_{.u})}{n(n-1)(k-1) S_{wst}^2}$$

$$2 \sum_{j < u} (\bar{y}_j - \bar{y})(\bar{y}_u - \bar{y}) = n(n-1)(K-1) S_{wst}^2 \rho_{wst}$$

put this above value in eq ①.

$$V(\bar{y}_{sys}) = \frac{1}{n^2 K} \left[n(K-1) S_{wst}^2 + n(n-1)(K-1) S_{wst}^2 \rho_{wst} \right]$$

$$= \frac{n(K-1) S_{wst}^2}{n^2 K} \left[1 + (n-1) \rho_{wst} \right]$$

$$= \frac{(K-1) S_{wst}^2}{N} \left[1 + (n-1) \rho_{wst} \right] \quad \left(\because K = \frac{N}{n} \right)$$

$$= \frac{\left(\frac{N}{n} - 1 \right) S_{wst}^2}{N} \left[1 + (n-1) \rho_{wst} \right]$$

$$V(\bar{y}_{sys}) = \frac{S_{wst}^2}{n} \left(\frac{N-n}{N} \right) \left[1 + (n-1) \rho_{wst} \right]$$

The systematic sample has the same precision as stratified random sample with one unit per stratum.

If $\rho_{wst} = 0$, no correlation

then

$$V(\bar{y}_{sys}) = \frac{(N-n)}{N \cdot n} S_{wst}^2$$

Comparison of Systematic Sampling with Simple Random Sampling

The performance of systematic sampling in relation to that of stratified or simple random sampling is greatly dependent on the properties of population. There are populations for which systematic sampling is extremely precise than simple random sampling. And for some other populations it is less precise than simple random sampling. For some populations and some values of n , $V(\bar{y}_{sys})$ may even increase when a larger sample is taken. Thus it is difficult to give general advice about the situations in which systematic sampling is to be recommended. A knowledge of the structure of the population is necessary for its most effective use.

This problem can be solved by two methods. One to compare different types of sampling on artificial populations in which Y_i is some simple

function of z .
The other is to make comparisons for natural populations.

Population In Random Order:-

Systematic Sampling is sometime used, for its convenience in populations in which the numbering of units is effectively random. If y_i is the variable of our interest.

For example: if a file is arranged alphabetically by surnames, if the item that is being measured has no relation to the surname of the individual. Then there will be no trend in y_i as we proceed along the file and no correlation between neighboring values.

In this situation we would expect systematic sampling to be essentially equivalent to simple random sampling.

THEOREM:-

If the variable y_i ($i=1, 2, 3, \dots, N$) are drawn at random from Super-population in which

$$E(y_i) = \mu, E(y_i - \mu)(y_j - \mu) = 0 \text{ for } i \neq j$$

$$E(y_i - \mu)^2 = \sigma_i^2$$

then $E(V_{sy}) = E(V_{rand})$.

Proof:-

For any specific finite population.

$$V_{rand} = \frac{(N-1)}{N} \frac{\sum_{i=1}^N (y_i - \bar{y})^2}{N-1}$$

$$\sum_{i=1}^N (y_i - \bar{y})^2 = \sum_{i=1}^N [y_i - \mu + \mu - \bar{y}]^2$$

$$= \sum_{i=1}^N [(y_i - \mu) - (\bar{y} - \mu)]^2$$

$$= \sum_{i=1}^N [(y_i - \mu)^2 + (\bar{y} - \mu)^2 - 2(y_i - \mu)(\bar{y} - \mu)]$$

$$= \sum_{i=1}^N (y_i - \mu)^2 + N(\bar{y} - \mu)^2 - 2(\bar{y} - \mu) \sum_{i=1}^N (y_i - \mu) \checkmark$$

$$= \sum_{i=1}^N (y_i - \mu)^2 + N(\bar{y} - \mu)^2 - 2N(\bar{y} - \mu)^2 \checkmark$$

$$\sum_{i=1}^N (y_i - \bar{y})^2 = \sum_{i=1}^N (y_i - \mu)^2 - N(\bar{y} - \mu)^2 \text{ --- (I) } \checkmark$$

Applying Expectation on both sides

$$E\left[\sum_{i=1}^N (y_i - \bar{y})^2\right] = E\left[\sum_{i=1}^N (y_i - \mu)^2\right] - N E(\bar{y} - \mu)^2 \rightarrow \text{Let only}$$

$$\text{As } E(y_i - \mu)^2 = \sigma_i^2$$

$$\sum_{i=1}^N E(y_i - \mu)^2 = \sum_{i=1}^N \sigma_i^2 \text{ --- (II)}$$

Let

$$\text{And } E(\bar{Y} - \mu)^2 = E\left[\frac{1}{N} \sum_{i=1}^N y_i - \mu\right]^2$$

$$= E\left[\frac{1}{N} \sum_{i=1}^N (y_i - \mu)\right]^2$$
$$= \frac{1}{N^2} E\left[\sum_{i=1}^N (y_i - \mu)^2 + \sum_{i \neq j} (y_i - \mu)(y_j - \mu)\right]$$

$$= \frac{1}{N^2} E\left[\sum_{i=1}^N (y_i - \mu)^2\right] + \sum_{i \neq j} E(y_i - \mu)(y_j - \mu)$$

Since i and j are not correlated as stated in the theorem, so

$$E(y_i - \mu)(y_j - \mu) = 0$$

So

$$E(\bar{Y} - \mu)^2 = \frac{1}{N^2} E\left[\sum_{i=1}^N (y_i - \mu)^2\right]$$
$$= \frac{1}{N^2} \cdot \sum_{i=1}^N E(y_i - \mu)^2$$

$$E(\bar{Y} - \mu)^2 = \frac{1}{N^2} \sum_{i=1}^N \sigma_i^2 \quad \text{--- (III)}$$

put E (II) and (III) in E (I), we have

$$E\left[\frac{1}{N} \sum_{i=1}^N (y_i - \bar{Y})^2\right] = \sum_{i=1}^N \sigma_i^2 - \frac{1}{N} \sum_{i=1}^N \sigma_i^2$$

$$= \left(1 - \frac{1}{N}\right) \sum_{i=1}^N \sigma_i^2$$
$$E\left[\frac{1}{N} \sum_{i=1}^N (y_i - \bar{Y})^2\right] = \left(\frac{N-1}{N}\right) \sum_{i=1}^N \sigma_i^2$$

Now

$$E(V_{\text{rand}}) = \frac{N-n}{N \cdot n} \cdot \frac{1}{N-1} \left(\frac{N-1}{N}\right) \sum_{i=1}^N \sigma_i^2$$

$$E(V_{\text{rand}}) = \frac{N-n}{n \cdot N} \sum_{i=1}^N \sigma_i^2 \quad \text{--- (A)}$$

Now Consider

Let $\bar{y}_u$ denote the mean of u th systematic sample.

For any specific finite population

$$V(\text{sys}) = \frac{1}{K} \sum_{u=1}^K (\bar{y}_u - \bar{Y})^2$$

$$= \frac{1}{K} \sum_{u=1}^K [\bar{y}_u - \mu + \mu - \bar{Y}]^2 \quad \begin{aligned} &\because \sum_u (\bar{y}_u - \mu) \\ &= K \left(\frac{\sum_u \bar{y}_u}{K} - \frac{K\mu}{K} \right) \\ &= K (\bar{Y} - \mu) \end{aligned}$$

$$= \frac{1}{K} \sum_{u=1}^K [(\bar{y}_u - \mu) - (\bar{Y} - \mu)]^2$$

$$= \frac{1}{K} \sum_{u=1}^K [(\bar{y}_u - \mu)^2 + (\bar{Y} - \mu)^2 - 2(\bar{y}_u - \mu)(\bar{Y} - \mu)]$$

$$= \frac{1}{K} \sum_{u=1}^K (\bar{y}_u - \mu)^2 + \frac{K}{K} (\bar{Y} - \mu)^2 - \frac{2}{K} (\bar{Y} - \mu) \sum_u (\bar{y}_u - \mu)$$

$$= \frac{1}{K} \sum_{u=1}^K (\bar{y}_u - \mu)^2 + (\bar{Y} - \mu)^2 - \frac{2}{K} \cdot K (\bar{Y} - \mu)^2$$

$$V(\text{sys}) = \frac{1}{K} \sum_{u=1}^K (\bar{y}_u - \mu)^2 - (\bar{Y} - \mu)^2$$

$$= \frac{1}{K} \sum_{u=1}^K \left[\frac{\sum_j y_{uj}}{n} - \mu \right]^2 - \left[\frac{\sum_i y_i}{N} - \mu \right]^2$$

$$= \frac{1}{nK} \sum_{u=1}^K \left[\frac{1}{n} \sum_j (y_{uj} - \mu)^2 \right] - \left[\frac{1}{N} \sum_{i=1}^N (y_i - \mu)^2 \right]$$

$$= \frac{1}{n^2 K} \sum_{u=1}^K \left[\sum_j (y_{uj} - \mu)^2 \right] - \frac{1}{N^2} \left[\sum_{i=1}^N (y_i - \mu)^2 \right]$$

Part I Part II

$$= \frac{1}{n^2 K} \sum_{u=1}^K \left[\sum_j (y_{uj} - \mu)^2 + \sum_{j \neq g} (y_{uj} - \mu)(y_{ug} - \mu) \right]$$

$$= -\frac{1}{N^2} \left[\sum_{i=1}^N (y_i - \mu)^2 + \sum_{i \neq j} (y_i - \mu)(y_j - \mu) \right]$$

$$= \frac{1}{n \cdot \frac{N}{n}} \sum_{u=1}^k \sum_{j=1}^n (y_{uj} - \mu)^2 + \sum_{u=1}^k \sum_{j \neq g} (y_{uj} - \mu)(y_{ug} - \mu)$$

$$= -\frac{1}{N^2} \sum_{i=1}^N (y_i - \mu)^2 - \frac{1}{N^2} \sum_{i \neq j} (y_i - \mu)(y_j - \mu)$$

$$= \frac{1}{nN} \sum_{i=1}^N (y_i - \mu)^2 + \sum_{u=1}^k \sum_{j \neq g} (y_{uj} - \mu)(y_{ug} - \mu)$$

$$- \frac{1}{N^2} \sum_{i=1}^N (y_i - \mu)^2 - \frac{1}{N^2} \sum_{i \neq j} (y_i - \mu)(y_j - \mu)$$

Applying Expectation on both sides.

$$E V_{sys} = \frac{1}{nN} \sum_{i=1}^N E (y_i - \mu)^2 + \sum_{u=1}^k \sum_{j \neq g} E (y_{uj} - \mu)(y_{ug} - \mu)$$

$$- \frac{1}{N^2} \sum_{i=1}^N E (y_i - \mu)^2 - \frac{1}{N^2} \sum_{i \neq j} E (y_i - \mu)(y_j - \mu)$$

$$E V_{sys} = \frac{1}{nN} \sum_{i=1}^N \sigma_i^2 + 0 - \frac{1}{N^2} \sum_{i=1}^N \sigma_i^2 - 0.$$

$$= \sum_{i=1}^N \sigma_i^2 \left[\frac{1}{nN} - \frac{1}{N^2} \right]$$

$$E V_{sys} = \sum_{i=1}^N \sigma_i^2 \left[\frac{N-n}{N^2 n} \right]$$

$$E V_{sys} = \frac{N-n}{N^2 n} \sum_{i=1}^N \sigma_i^2 \quad \text{--- (B)}$$

From Eq (A) & (B),

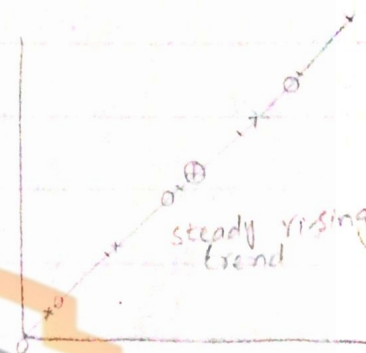
$$E V_{rand} = E V_{sys}. \text{ As reqd.}$$

Population follows a linear trend:-

If the population follows a linear trend then V_{sy} and V_{st} both are smaller than V_{grand} , and V_{gst} is smaller than V_{jsy} . Therefore V_{gst} will be most precise.

X = Systematic Sample

O = Stratified Random Sample



Because the population follows linear trend, and its trend is rising steadily. We suppose that population is in some order. By systematic sampling to select a sample of n units, we take a unit at random from first k units and every k th unit thereafter.

In the figure, selected units are denoted by 'x'. In the figure we see that the first unit is selected at the start of first of ' k ' units. Now from every ' k ' units we select a unit from the start of k th one. If first selected unit is not right, then it would effect on the whole sample. We can not get precise results. Also these

units are selected from the beginning of every k units. Therefore the sample would not represent any of the k units.

While when we select a unit from every ' k ' units at random, then we see that a unit is selected from every ' k ' units from different places. From first k units in the middle in the second k units it is selected from last and so on. This selected sample would give more precise results than sample selected by systematic random sampling.

THEOREM:-

If the population follows a linear trend then show that

$$V_{st} \leq V_{sys} \leq V_{rand}.$$

Proof:-

We know that

$$V_{ran} = \left(\frac{N-n}{N \cdot n} \right) S^2$$

$$S^2 = \sum_i^N \frac{(Y_i - \bar{Y})^2}{N-1} = \frac{1}{N-1} \left[\sum Y_i^2 - \frac{(\sum Y_i)^2}{N} \right]$$

If the population follows a linear trend the value $Y_i = i = 1, 2, 3, \dots, N$

$$\begin{aligned} \sum_i^N Y_i &= \sum_i^N i = 1 + 2 + 3 + \dots + N = \frac{N(N+1)}{2} \\ \sum_i^N Y_i^2 &= \sum_i^N i^2 = 1^2 + 2^2 + 3^2 + \dots + N^2 \\ &= \frac{N(N+1)(2N+1)}{6} \end{aligned}$$

Then

$$\begin{aligned} S^2 &= \frac{1}{N-1} \left[\frac{N(N+1)(2N+1)}{6} - \frac{1}{N} \left(\frac{N(N+1)}{2} \right)^2 \right] \\ &= \frac{N(N+1)}{(N-1)} \left[\frac{2N+1}{6} - \frac{N+1}{4} \right] \\ &= \frac{N(N+1)}{(N-1)} \left[\frac{4N+2 - 3N-3}{12} \right] = \frac{N(N+1)}{(N-1)} \left(\frac{N-1}{12} \right) \end{aligned}$$

$$S^2 = \frac{N(N+1)}{12} \quad \text{--- (A)}$$

$$V_{\text{rand}} = \frac{N-n}{N \cdot n} \cdot \frac{N(N+1)}{12}$$

$$V_{\text{rand}} = \left(\frac{N}{n} - 1 \right) \left(\frac{N+1}{12} \right) = \frac{(K-1)(N+1)}{12} \quad \text{--- (I)}$$

$$\text{Now } V_{\text{sys}} = \frac{1}{K} \sum_{u=1}^K (\bar{y}_u - \bar{Y})^2 = \frac{1}{K} \sum_{u=1}^K \left[\bar{y}_u^2 - \frac{(\sum \bar{y}_u)^2}{K} \right]$$

For systematic Sampling the mean of 2nd sample exceeds that of the first

by 1, the mean of the 3rd that exceeds that of the 2nd by 1 and so on.

Thus the mean $\bar{y}_u$ may be replaced by the no's $1, 2, 3, \dots, k$

Hence

$$\sum_{i=1}^k \bar{y}_i = \sum_{i=1}^k i = 1+2+3+\dots+k$$

$$= \frac{k(k+1)}{2}$$

$$\sum_{i=1}^k \bar{y}_i^2 = \sum_{i=1}^k i^2 = 1^2+2^2+3^2+\dots+k^2 = \frac{k(k+1)(2k+1)}{6}$$

$$V_{sys} = \frac{1}{k} \left[\frac{k(k+1)(2k+1)}{6} - \frac{1}{k} \left(\frac{k(k+1)}{2} \right)^2 \right]$$

$$= \left[\frac{k(k+1)(2k+1)}{6} - \frac{1}{k} \frac{k^2(k+1)^2}{4} \right]$$

$$= \frac{k(k+1)}{k} \left[\frac{2k+1}{6} - \frac{k+1}{4} \right]$$

$$= (k+1) \left[\frac{4k+2-3k-3}{12} \right]$$

$$V_{sys} = \frac{(k+1)(k-1)}{12} \quad \text{--- (II)}$$

And $V_{\bar{y}_{st}} = \frac{N-n}{N \cdot n} S_w^2$

If sample allocation is proportional N_h , and within strat variances

are equal for all stratas and then it is denoted by S_w^2 .
 Assuming $y_i = i = 1, 2, 3, \dots, k$ for k th stratum.
 So to find S_w^2 we need only replace 'N' by 'k' in expression (A).
 Thus

$$V_{st} = \frac{N-n}{N} S_w^2 = \frac{k(k+1)}{12}$$

$$\text{So } V_{st} = \frac{N-n}{N} \cdot \frac{k(k+1)}{12}$$

$$\therefore N = nk$$

$$k = \frac{N}{n}$$

$$= \frac{nk - n}{nk} \cdot \frac{k(k+1)}{12}$$

$$= \frac{n(k-1)}{n^2} \cdot \left(\frac{k+1}{12} \right)$$

$$V_{st} = \frac{(k-1)(k+1)}{12n}$$

$$V_{st} = \frac{k^2 - 1}{12n} \quad \text{--- (III)}$$

Comparing eq's (I), (II) and (III).

$$V_{sys} \leq V_{ran}$$

Comparing eq's (II) and (III)

$$V_{st} \leq V_{sys}$$

So finally

$$V_{st} \leq V_{sys} \leq V_{ran}$$

$$\frac{k^2 - 1}{12n} \leq \frac{k^2 - 1}{12} \leq \frac{(k-1)(k+1)}{12}$$

Equality occurs when $n=1$. then

$$\frac{k^2-1}{12(1)-1} = \frac{k^2-1}{12} = \frac{(k^2-1)}{12}$$

$$\because N=nk \\ N=k$$

$$\frac{k^2-1}{12} = \frac{k^2-1}{12} = \frac{k^2-1}{12}$$

Thus for removing the effect of a linear trend suspected and unsuspected, the Systematic Sample is much more effective than a Simple Random Sample but less effective than the Stratified Sampling.

Theorem

if loss function due to $\bar{y}$ for not being equal to $\bar{Y}$ is $d|\bar{y}-\bar{Y}|$ and the cost function is $C = C_0 + C_1 n$ show that the sample size in case of S.R.S ignoring F.P.C is $\left(\frac{dS}{C_1 \sqrt{2\pi}}\right)^{2/3}$

Sol

by definition $L(n) = d|\bar{y}-\bar{Y}|$

$$L(n) = E[L(n)] = E[d|\bar{y}-\bar{Y}|]$$

$$= d E(|\bar{y}-\bar{Y}|)$$

$$= d M.D(\bar{y}) \rightarrow (1)$$

Now we know that the sampling distribution of $\bar{y}$ is normally distributed with mean $\bar{y}$ and variance σ^2/n i.e.

$$\text{Also } M.D(\bar{y}) = \frac{4}{5} S.D(\bar{y})$$

$$= \frac{4}{5} \sqrt{\frac{1-f}{n}} S \quad \left| \frac{4}{5} = \frac{\sqrt{2}}{\sqrt{n}} \right.$$

$$= \sqrt{\frac{2}{n}} \sqrt{\frac{1-f}{n}} S$$

ignoring f.p.c

$$= \sqrt{\frac{2}{n}} \frac{S}{\sqrt{n}} = \sqrt{\frac{2}{n}} S$$

$$\text{hence } M.D(\bar{y}) = \sqrt{\frac{2}{n}} S$$

put in Eq (A)

$$L(n) = d S \sqrt{\frac{2}{n}}$$

Loss function = Expected loss + Cost function

$$= d \sqrt{\frac{2}{n}} S + C_0 + C_1 n$$

$$L'(n) = \frac{dL(n)}{dn} = -\frac{dS}{n^{3/2} \sqrt{2}} + C_1$$

$$= \frac{-dS}{n^{3/2} \sqrt{2}} + C_1$$

$$\text{put } L'(n) = 0$$

$$-\frac{dS}{n^{3/2} \sqrt{2}} + C_1 = 0$$

$$C_1 = \frac{dS}{n^{3/2} \sqrt{2}}$$

$$n^{3/2} = \frac{dS}{C_1 \sqrt{2}}$$

$$\Rightarrow n = \left(\frac{dS}{C_1 \sqrt{2}} \right)^{2/3}$$

if we are $d(\bar{y} - \bar{y})^2$

$$\begin{aligned} \text{then } L_n &= E(l_n) \\ &= E d(\bar{y} - \bar{y})^2 \\ &= d E(\bar{y} - \bar{y})^2 \end{aligned}$$

$$= d v(\bar{y})$$

$$= d \frac{1}{n} s^2 \frac{d s^2 (1-f)}{n}$$

$$L(n) = \frac{d s^2}{n} \text{ if F.P.c ignored}$$

$$\text{Loss function} = E(l_n) + E(\text{cost})$$

$$L(n) = \frac{d s^2}{n} + C_0 + C_1 n$$

$$\dot{L}(n) = -\frac{d s^2}{n^2} + C_1$$

$$-\frac{d s^2}{n^2} \neq C_1 = 0$$

$$C_1 = \frac{d s^2}{n^2}$$

$$n^2 = \frac{d s^2}{C_1}$$

$$n = \left(\frac{d s^2}{C_1} \right)^{1/2}$$

$$n = \sqrt{\frac{d s^2}{C_1}}$$

Design Effect

The ratio of the variance estimate obtained from the complex sampling procedure to the variance of the estimate obtained from SRS of same number of units. This concept is introduced by Kish in 1965.

USE of Design effect:

The design effect has two primary uses in sample size estimation and in appraising the efficiency of more complex problems. For instance, in estimating the proportion of people who possess some attribute it is often convenient to use household instead of person as a sampling unit when the sampling fraction ~~unit~~ ^{units} are small we can therefore, estimate sample size by calculating the 'n' needed with SRS sample of persons.

Def-
$$\frac{\text{variance from complex sampling}}{\text{variance from SRS}}$$